Does the snRNA-seq evidence for hepatocyte de-zonation survive matched sampling?

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Abstract. Gribben et al. report that during human chronic liver disease (MASLD) hepatocytes progressively lose their spatial zonation and acquire cholangiocyte-like (biliary) features. We re-examined the single-nucleus RNA-seq evidence for the *progressive de-zonation* part of that claim. The complication we found is structural: **disease stage is collinear with how the tissue was obtained.** Healthy and end-stage samples are deceased-donor organ tissue (a $\sim$1 cm^3 cube / explant); the F0–F4 disease spectrum is 16-gauge needle biopsy. So "healthy $\rightarrow$ MASLD $\rightarrow$ end-stage" is also "organ cube $\rightarrow$ needle $\rightarrow$ explant," and a single pooled correlation cannot tell disease from acquisition. After setting aside the source-confounded endpoints and re-testing on the matched needle-biopsy axis — using raw UMI counts, depth-normalized per nucleus, with the **donor** ($\sim$47) as the unit of inference — we find **no transcriptional de-zonation signal across fibrosis F1$\rightarrow$cirrhotic F4**: cells keep their pericentral/periportal identity, and a large coordinated shift is affirmatively excluded. A per-gene pseudobulk differential-expression scan finds no zonation gene moving (GLUL F4-vs-F1 FDR 0.80); its one per-gene signal is a small **biliary/ductular-marker rise** at F4, more consistent with cholangiocyte ambient RNA plus rare mixed/doublet nuclei (the ductular reaction) than with broad hepatocyte transdifferentiation. Pre-specified gene-set tests (CAMERA/ROAST/GSEA) plus a per-cell measure then reveal the one real biopsy-internal change: the **pericentral detox program dims** — detox transcripts within pericentral nuclei decline across fibrosis (donor-level Spearman $\rho = -0.48$, $p = 0.003$), a *relative* decline in level while the cells keep their identity, i.e. functional dimming, not de-zonation. We do not claim to overturn the paper; biliary source attribution is **a lead, not closed.**

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1 Question and scope

The published claim has two parts: hepatocytes progressively *lose zonation*, and they acquire *cholangiocyte-like plasticity*. The paper tests de-zonation with a marker–marker correlation (Welch's t-test) pooled across all 47 donors, and confirms plasticity with tissue imaging.

Our question is narrow: **does the single-nucleus transcriptional evidence for progressive de-zonation survive a matched-source analysis?** Scope: raw snRNA counts only. We do not touch the paper's imaging, immunostaining, protein, organoid, or spatial-architecture evidence — and we agree with the paper that its strong signal is an end-stage phenomenon. We are not claiming to overturn the full paper.

Zonation means hepatocytes run different programs by lobule position: pericentral (PC, near the central vein) cells favor *GLUL, CYP3A4* and drug-metabolizing enzymes; periportal (PP) cells favor *ASS1, PCK1, HAL, ALDOB* (urea cycle, gluconeogenesis). De-zonation could break this in several distinct ways — PC cells deplete, zonal genes turn off, cells co-express both programs, the gradient compresses, or the PP:PC balance shifts. Because each failure mode has a different count signature, we tested them separately rather than leaning on one global score.

2 Data and the key confound

The dataset is not one homogeneous disease series. Tissue source is tied to stage:

Group	Tissue source	Acquisition
Healthy (n=4)	Deceased-donor organ tissue (one with biliary-stone disease)	$\sim$1 cm^3 organ cube
Disease F0–F4	MASLD/NASH/cirrhosis	16-gauge end-cut needle biopsy
End-stage (n=5)	Explanted cirrhotic organs (left/right/caudate lobes)	Whole-organ explant

Table 1: The three sampling groups. Both ends of the apparent disease axis are organ tissue; only the F0–F4 middle is needle biopsy.

Both *ends* of the apparent disease axis are organ tissue; only the F0–F4 middle is needle biopsy (grounded directly in Paper 1's Methods, `papers/s41586-024-07465-2.pdf`; see `reports/SYNTHESIS.md` §1a). The trajectory "healthy → MASLD → end-stage" is therefore also an **acquisition trajectory** — ischemia, procurement delay, dissociation, organ-cube-vs-needle, and sequencing batch all move together with stage (run almost perfectly predicts tissue source: Cramér's V $\approx$ 0.84). This is the load-bearing issue: it motivates excluding the ends and re-testing biopsy-only.

Figure 1 — Study design and the source confound *(schematic; see presentation slide 3)*: a three-group panel showing that disease stage runs collinear with acquisition mode, so a pooled comparison reads a procurement discontinuity as biology. The three sampling groups are laid out in Table 1.

3 Methods that matter

Only the choices that explain why our result differs.

3.1 Raw counts, not the shipped matrix. The distributed matrix is an SCT model output (depth squeezed); we extracted real RNA-assay UMI counts. All molecule inference is on raw counts.

3.2 Depth-normalized anchor classification. Each hepatocyte nucleus is down-thinned to a common 1,500-UMI budget (binomial thinning, averaged over draws) so detection reflects biology not depth. A marker is "on" at ≥ 2 UMI (ambient-robust). Using strict mutually-exclusive anchors — PC = *GLUL/CYP3A4*, PP = *ASS1/PCK1/HAL/ALDOB* — each nucleus is labeled PC, PP, dual, or null. The unit of inference is the **donor**, never the cell. (This is differential-abundance / composition analysis; we call it "anchor classification.")

3.3 Donor-level pseudobulk DGE. Sum each donor's hepatocyte raw UMIs into one count profile; run edgeR (TMM normalization, negative-binomial quasi-likelihood), primary contrast F4-vs-F1 across 38 biopsy donors. Purpose: a genome-wide check for **non-zonation** programs given that zonation structure is treated as settled — not a second de-zonation test.

4 Results

4.1 R1 — The endpoints carry acquisition stress, so we exclude them

A curated stress module (immediate-early + heat-shock genes), depth-normalized, runs at **0.074** mean UMIs/nucleus in biopsies, **0.282** in healthy organ cubes, and **1.675** in end-stage explants (∼22× biopsy). The decisive control is cross-lineage: end-stage-vs-biopsy stress is ∼18.5× in hepatocytes and ∼18.2× in **endothelial cells**, which carry no zonation program. A signal identical in a non-zonated lineage is whole-organ procurement handling, not hepatocyte biology — and it perturbs exactly the oxygen/Wnt-gradient axis the de-zonation claim rests on. This justifies excluding the source-confounded endpoints from the disease axis.

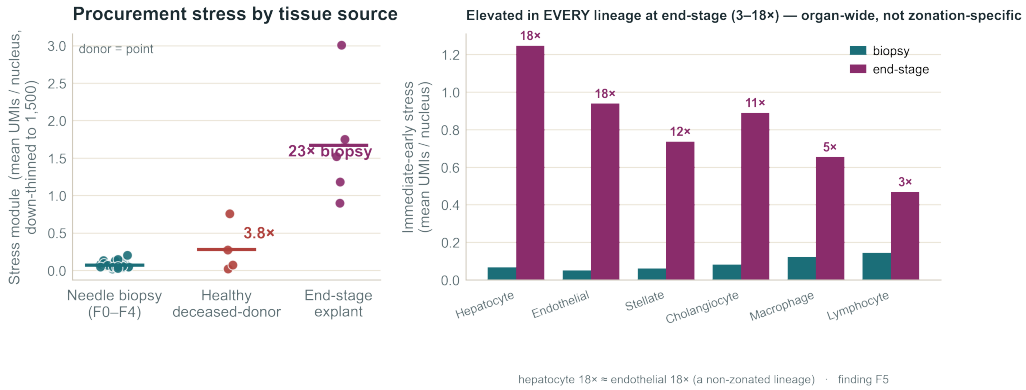


Figure 1: Procurement stress by group and lineage. The endpoints (organ cube, explant) carry an organ-wide acute stress signature; biopsies are uniformly low and flat.

4.2 R2 — On the matched biopsy axis, there is no de-zonation signal

Across F0–F4 needle biopsies every cell-redistribution failure mode comes up flat:

- **Depletion:** PC-anchor fraction 36/19/23/22/21% (F0–F4) — flat/non-monotone.
- **Co-expression:** the dual fraction looks inflated (∼7–10%) only at a 1-UMI threshold; at the ambient-robust ≥ 2-UMI cut it collapses to ∼**0.2–0.6%** with no trend — the 1-UMI signal was ambient soup.
- **Turn-off:** null fraction flat.
- **Composition:** PP:PC ratio 0.62/1.16/1.01/1.10/1.18 — flat/non-monotone.

The poles stay dominant; at most a mild non-monotone middle-ward drift. This is no de-zonation *signal* — not a positive proof that zonation is "preserved," only that no de-zonation is detectable above the donor-level scatter. The null is robust across the full sensitivity grid and a marker-set ladder from 2 to 1,637 genes (including Paper 2's own 40 landmark genes). A per-cell view of the same data (the pericentral↔periportal balance of each nucleus) tells the same story: across biopsy F1→F4 the cells stay spread along the gradient, with only the confounded explant collapsing to a single pole.

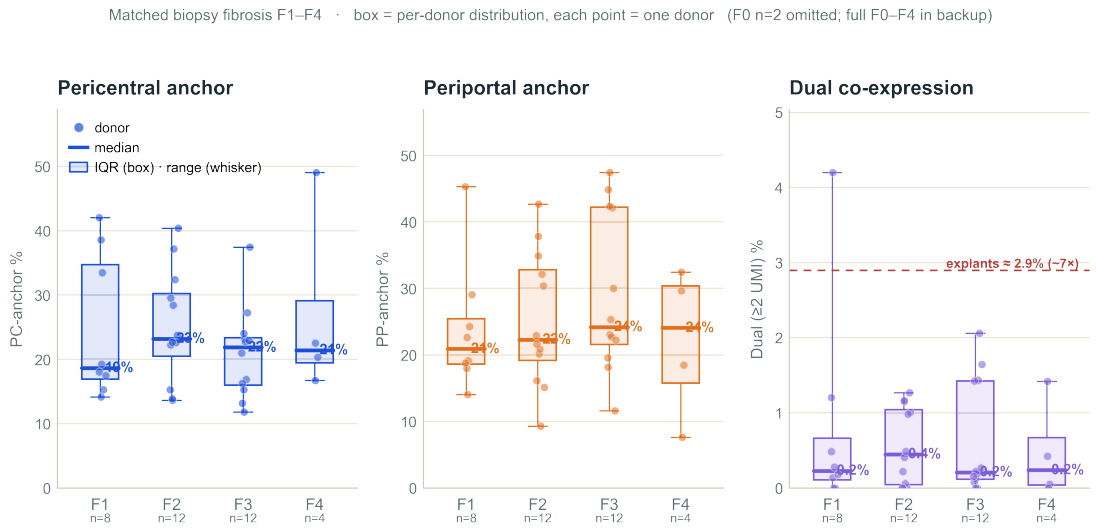


Figure 2: Matched-biopsy anchors (3-panel): pericentral and periportal anchor fractions flat across fibrosis F1–F4, and the dual co-expression fraction near-zero (∼0.4%, vs ∼2.9% in explants) at the ≥ 2-UMI cut — the apparent 1-UMI co-expression was ambient soup. Box = per-donor spread; each point = one donor.

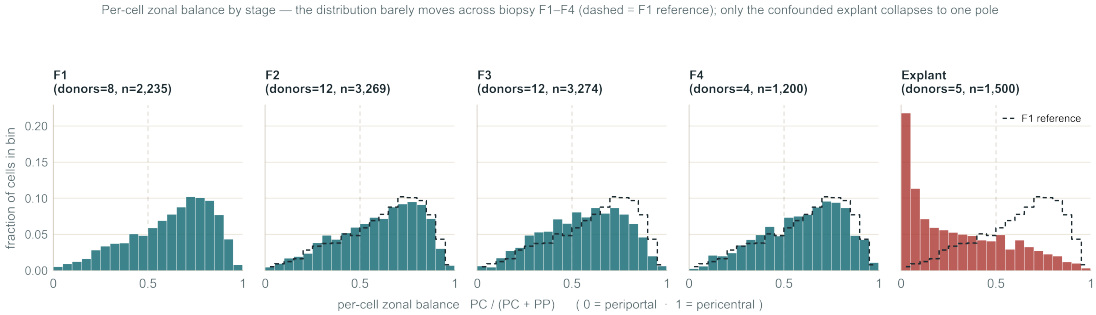


Figure 3: Per-cell zonal balance by stage (down-thinned to 1,500 UMIs). The x-axis is each nucleus's pericentral share PC/(PC+PP): spread = gradient present, one pile = collapse. The dashed outline repeats the F1 distribution on every panel: biopsy F1–F4 barely move; only the explant collapses to the periportal pole. (Null and PP:PC ratio: `fig_secondary.png`.)

4.3 R3 — Robustness and an honest bound

The result is stable across the sensitivity grid (depth budgets 1,000/1,500/3,000; GLUL-only/CYP3A4-only/both; Monte-Carlo SD on donor medians 0.006–0.010). An affirmative equivalence test (TOST, 90% CI, donor-level anchor fractions) puts the F4-vs-F1 PC-anchor change at **+0.024 (90% CI** −0.147 **to** +0.194**)**. So we **exclude a large shift on the order of 20 percentage points** (the well-powered F1→F3 interior tightens to ∼ ±0.12); we do **not** exclude a subtle drift of ≤ 10 points (donor SD is large at F4 n=4). Large re-zonation is ruled out; subtle drift is not.

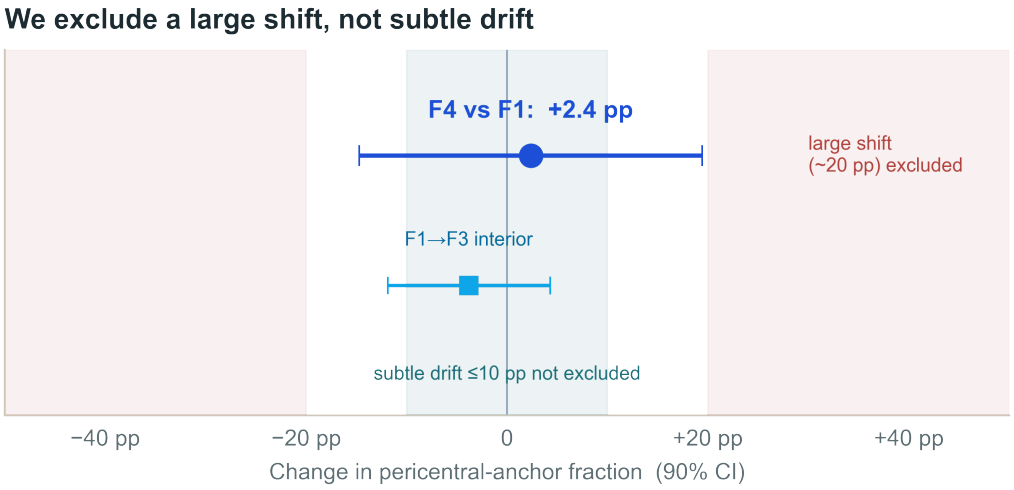


Figure 4: Equivalence bound on the F4-vs-F1 pericentral-anchor change: a large shift is excluded, a subtle one is not.

4.4 R4 — The only per-gene DGE signal is a biliary rise, likely compositional

Genome-wide, of ∼21,000 genes only **64** reach FDR < 0.05 at F4-vs-F1, and they are a biliary/ductular set — *EPCAM, GRHL2, SPINT2, SOX4, SOX9, B3GNT3* — plus *CXCL10*. Zonation genes are flat at the expression level (**GLUL F4-vs-F1 FDR 0.80 — well above the 0.05 cutoff, so not significant**; all detox genes FDR > 0.79), an independent count-based check agreeing with the anchor classification. So the per-gene discovery scan finds **a biliary-marker rise inside hepatocyte-labeled pseudobulk, not zonation loss.**

Source attribution (a side lead): the headline genes are 5–78× more abundant in cholangiocytes than hepatocytes; the cholangiocyte fraction rises to ∼8% at F4 (the ductular reaction); only ∼0.4% of hepatocyte nuclei robustly co-express ≥ 2 biliary markers (a prevalence over *all* hepatocytes — it argues against *mass* transdifferentiation but does not exclude a rare sub-population); ambient correction (decontX) halves the hits 64→34 and drops *SOX4/SOX9* below significance. But *EPCAM/B3GNT3/SPINT2* survive (decontX is a consistency check, not a verdict, so a rare intrinsic biliary state is not excluded), and *CXCL10* is separate — its hepatocyte expression does not track the cholangiocyte fraction (corr -0.09), so it is a candidate real inflammatory signal, not ambient. Careful conclusion: the biliary signal is **real in hepatocyte-labeled pseudobulk, source not resolved**, and the weight of evidence favors **ambient RNA + rare mixed/doublet nuclei** over broad transdifferentiation — a lead, not a closed verdict.

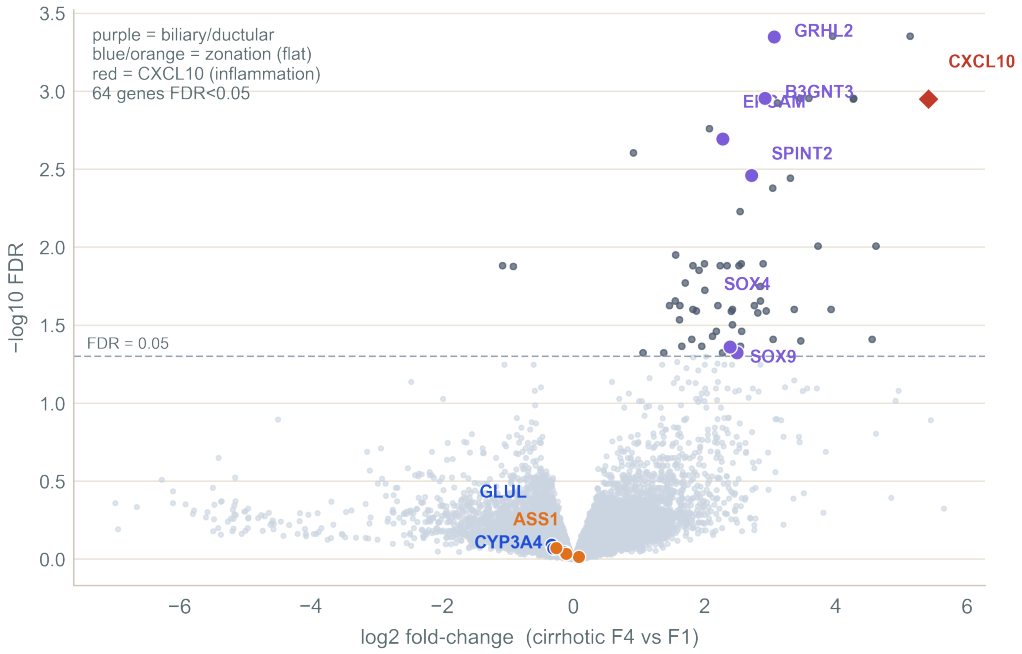


Figure 5: Volcano (F4-vs-F1): zonation genes flat; the highlighted hits are biliary/ductular markers. Compositional audit places these genes as cholangiocyte-dominant. (`fig_volcano.png` + `fig_compositional.png`.)

4.5 R5 — The one real change: the pericentral detox program dims

A per-gene scan is blind to a weak *coordinated* program — many genes each drifting a little in the same direction, none crossing FDR alone. Pre-specified gene-set tests are built for exactly that. Over ten sets fixed in advance (pericentral identity and detox, periportal identity and urea cycle, bile-acid/lipid, cholangiocyte, plus interferon / fibrogenesis / ER-stress *controls* whose answers biology already dictates), three independent methods agree: `camera` (competitive), `mroast` (self-contained), and GSEA. The **pericentral detox programs go down** with fibrosis — phase-I CYP detox (camera FDR 2×10^{-6}), pericentral identity (Paper 2 landmarks, FDR 1.5×10^{-4}), phase-II conjugation (FDR 7×10^{-3}) — while the controls behave exactly as they must (fibrogenesis and inflammation up, ER-stress flat), which is what tells us the test is reading real biology, not noise.

Because a competitive test on relative data could be fooled by other genes inflating (a "pie-slice" artifact), we checked adversarially: the result survives composition-robust TMM normalization, survives the self-contained `mroast` test (the CYP detox core clears alone), survives dropping the up-going genes — and, decisively, a **per-cell measure** that composition cannot touch confirms it. Counting detox transcripts per pericentral nucleus after equalizing read depth (down-thinning every nucleus to 1,500 UMIs), the level falls steadily across fibrosis: ∼11.9 (F1) to ∼8.8 (F4) transcripts per nucleus, a donor-level Spearman $\rho = -0.48$ ($p = 0.003$).

This is **dimming**, not de-zonation. Like a radio: the station did not change genre (the cells keep their pericentral identity — the anchor fractions in R2 stay flat), but the volume dropped (the detox program's level falls). It works precisely because the identity anchors (GLUL, CYP3A4 — a yes/no detection call) are *different genes* from the detox-module genes measured for dimming (CYP2E1, CYP1A2, ADH4…), so the anchor fraction can stay flat while the module quiets. Three honest caveats: this is a **relative** decline (detox transcripts within a fixed sequencing budget, not proven absolute molecule loss); it is a donor-level trend; and it could partly reflect a shift among pericentral subzones rather than pure within-cell downregulation. It does *not* rescue the paper's identity-loss claim.

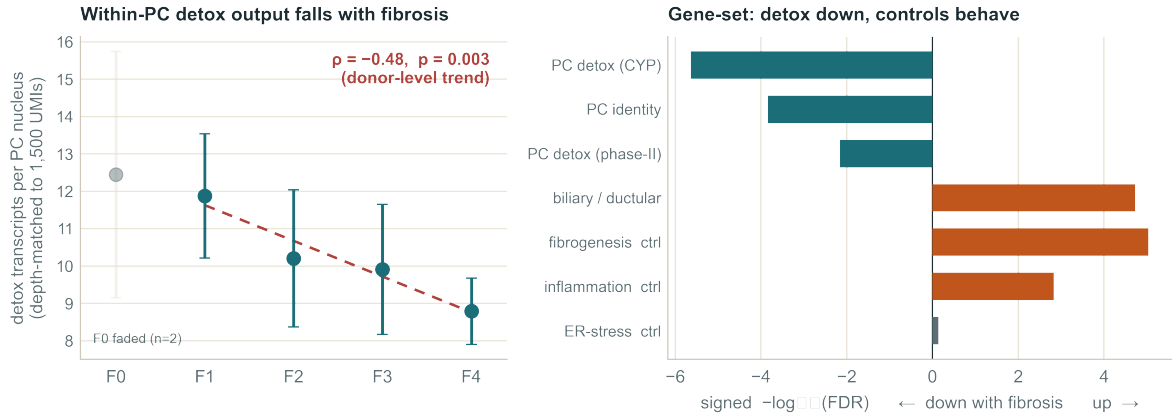


Figure 6: The pericentral detox module dims. *Left:* detox transcripts per pericentral nucleus (depth-matched to 1,500 UMIs) fall from ∼11.9 (F1) to ∼8.8 (F4); donor-level Spearman $\rho = -0.48$, $p = 0.003$. *Right:* pre-specified gene-set programs — pericentral detox and identity down, biliary and fibrogenesis/inflammation controls up, ER-stress flat. Measured on detox genes disjoint from the GLUL/CYP3A4 identity anchors, so the anchor fraction stays flat while the module dims.

5 Interpretation

1. **The full trajectory is source-confounded** — do not read healthy → end-stage as a clean continuation of the disease axis. The endpoints are organ tissue carrying organ-wide procurement stress.
2. **In matched biopsies, there is no transcriptional de-zonation signal** — every cell-redistribution failure mode (depletion, co-expression, turn-off, composition shift) is flat across F1→cirrhotic F4, and a large coordinated shift is affirmatively excluded. The cells keep their zonal identity.
3. **The one real biopsy-internal change is dimming, not de-zonation** — the pericentral detox program declines in *level* (gene-set down-regulation plus a per-cell depth-matched trend, $\rho = -0.48$, $p = 0.003$) while the cells retain their pericentral identity. This is a relative decline in a zonated program's output, not a loss of identity or a transdifferentiation.
4. **The F4 biliary/ductular signal is real but source-uncertain** — most consistent with cholangiocyte ambient RNA plus rare mixed/doublet nuclei, with *CXCL10* a possible real hepatocyte inflammatory exception. This source-attribution leg is a promising lead, not a closed finding.

6 Limitations and next steps

Stated plainly, not apologetically. This is a hackathon-scale re-analysis: no wet-lab, no imaging or spatial re-analysis, single cohort. The F4 biopsy stratum is thin (n=4), so subtle effects there are underpowered — which is why we lead the dimming result with the fibrosis-wide trend, not the F1-vs-F4 endpoint. The detox-dimming finding itself is a *relative* decline within a fixed sequencing budget (not proven absolute molecule loss), a donor-level trend, and could partly reflect a shift among pericentral subzones; in the strictly self-contained gene-set test only the CYP detox core clears alone. decontX is a sensitivity check using the paper's own labels (mild circularity), not proof of ambient origin. And anchor markers simplify what is really a continuous gradient; snRNA measures per-cell transcriptional balance, not lobule geometry — we report the absence of a transcriptional de-zonation *signal*, not spatial/architectural preservation.

Next steps: validate the detox-dimming gradient in spatial / independent-biopsy tissue and with a functional readout of pericentral detox capacity (and settle within-cell downregulation vs pericentral-subzone shift); leave-one-F4-donor-out DGE on the biliary hits; a quantitative contamination model asking whether the ambient/doublet load numerically explains the counts; and an independent MASLD

biopsy cohort. (The gene-set/pathway tests — CAMERA, ROAST, GSEA — that close the weak-coordinated-program gap are reported in R5.)

Appendix / backup

- **Why we abandoned the relative ruler.** An early z-scored "zonation coordinate" produced an apparent collapse. It was abandoned for two reasons: (a) it pools across the source confound, so a between-group acquisition discontinuity reads as within-group de-zonation (a Simpson's-paradox reversal — each healthy donor is individually zonated, anti-corr ≈ -0.21, yet pooling flips the sign to $\approx +0.01$); and (b) it is intrinsically depth-sensitive and mechanism-blind — turn-off, co-expression, and noise all shrink its spread alike. The count anchor classification separates the mechanisms; the ruler cannot. (`findings/relative_ruler_postmortem/`.)
- **End-stage is not one program.** Across the five explants the PC-anchor fraction ranges 3%→50% and PP:PC 0.13→20 (CL104 retains pericentral; CL16 collapses to periportal; CL18 floods into co-expression). Pooling five discordant, separately-procured organs into one correlation manufactures a uniform "collapse." (`reports/SYNTHESIS.md` §1f.)
- **Provenance.** Stress and per-donor maps: `load_bearing.py`; cross-lineage stress: `supplementary_checks.py`; marker-set invariance: `all_sets_census.py`; genome-wide DGE: `dge_genomewide.py` / Plan A `findings/dge_plan_a/`; equivalence bound: `findings/equivalence_bound/`. Full numeric record: `FINDINGS.md` and `CLAIMS_LEDGER.md`.